

COMPUTER GRAPHICS PROJECT

Bloxorz 3D

Three.js / WebGL puzzle game

SCOPE

What we built

8 verified levels

rolling cuboid

fragile tiles

heavy and soft switches

bridges

teleport split mode

level editor

saved progress

hint solver

GAME STATE

Discrete first

```
state = x, z, orientation
```

```
orientation =  
    standing  
    lying_x  
    lying_z
```

```
occupied cells decide:  
    legal move  
    fall  
    switch  
    goal
```

MOVEMENT

Animation follows the rule

input changes grid state

mesh position is interpolated

mesh rotation uses quaternion motion

after each move:

 snap back to exact grid coordinate

Rules on the board

normal tile	supports the block
fragile tile	breaks under standing block
heavy switch	standing block only
soft switch	any occupied cell
bridge	toggled tile
teleport	split into two cubes
goal	standing block only

Course connection

geometry

model transform

view transform

projection

rasterization

fragment shading

frame buffer

RENDERING

Scene setup

PerspectiveCamera

MeshPhysicalMaterial

DirectionalLight

AmbientLight

shadows

fog

SMAA

bloom

vignette

optional GPU path tracing

INTERACTION

Player input

keyboard controls

touch controls

third-person camera

move counter

par display

hints

localStorage progress

SOLVER

BFS over puzzle states

state key:

x, z

orientation

bridge mask

split status

verified par:

1	First Steps	8
2	Over the Edge	7
3	Fragile Ground	6
4	Open Sesame	6
5	Heavy Duty	12
6	Split Decision	8
7	Twin Bridges	13
8	The Gauntlet	15

Files to explain

src/main.js	game flow
src/grid.js	board, tiles, bridges
src/controls.js	movement and validation
src/levels.js	levels, pars, themes
src/solver.js	hint search
src/editor.js	custom levels
src/scene.js	renderer, camera, lights
src/particles.js	visual feedback
scripts/check-levels.js	level verification

Five minutes

1. level select and saved progress
2. Level 1 movement and goal
3. fragile tile
4. switch and bridge
5. teleport split and merge
6. hint
7. editor if time remains